Activity A7 - Evaluate Performance and Conduct an Internal Audit

Course: Fundamentals of ISO 14001 Environmental Management System (EMS) (TGS-2023037592) Ability: A5, A6 Objective: Evaluate environmental performance, compliance evidence and EMS effectiveness through risk-based audit sampling. Mode: Team analysis with individual evidence submission Suggested time: 60 minutes

Scenario

MEP reports that waste diversion improved, but production volume changed and two months of wastewater data are missing. The prior audit programme covered every department equally and did not state objectives.

Meridian EcoPack Pte Ltd (MEP) is a fictional Singapore SME with 120 employees producing printed corrugated and fibre-based packaging for regional food, electronics and healthcare customers. Two key customers require evidence of an ISO 14001:2026-aligned EMS and stronger value-chain environmental controls before contract renewal.

Task and expected output

Analyse MEP's six-month KPI data, sample control records and conduct a process-based audit before management review.

Produce: A KPI interpretation, compliance-evaluation record, audit plan, evidence log and defensible findings.

Before you begin

- Open a working copy of the Excel workbook; do not overwrite the supplied evidence.
- Allocate a facilitator, evidence recorder, checker and presenter.
- Read every evidence card before making a compliance or significance conclusion.
- If the activity concerns an incident, this is a tabletop exercise: do not handle chemicals or simulate unsafe physical actions.

Decision questions

1. What can the data support, what is uncertain and what requires normalisation or validation?
2. How is evaluation of compliance different from routine monitoring and from internal audit?
3. Does each finding state the criterion, objective evidence and gap without prescribing a solution?

Detailed procedure

Step 1 - Verify KPI definitions, boundaries, units, calibration/data-quality controls and missing-data treatment

Verify KPI definitions, boundaries, units, calibration/data-quality controls and missing-data treatment.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 2 - Normalise energy, water and waste measures to production where appropriate and interpret trends against thresholds

Normalise energy, water and waste measures to production where appropriate and interpret trends against thresholds.

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Step 3 - Evaluate a sample of obligations using current evidence and record status, basis, evaluator and action

Evaluate a sample of obligations using current evidence and record status, basis, evaluator and action.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 4 - Set audit objectives, scope, criteria, methods, timing and independence using risk and prior performance

Set audit objectives, scope, criteria, methods, timing and independence using risk and prior performance.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 5 - Follow one aspect from policy and planning through operation, monitoring, competence and improvement evidence

Follow one aspect from policy and planning through operation, monitoring, competence and improvement evidence.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 6 - Triangulate interview, observation and document evidence; write findings and conduct a concise closing meeting

Triangulate interview, observation and document evidence; write findings and conduct a concise closing meeting.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Step 7 - Prepare management-review inputs and decisions needed from the combined evidence

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Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

Evidence to submit

- Completed Activity-A7-Workbook.xlsx.
- One evidence-backed team decision and one documented peer challenge.
- Any assumption or missing source clearly labelled for follow-up.

Acceptance checklist

[ ] Conclusions respect data limitations; compliance status is evidence-based; the audit trail is reproducible; every finding contains criterion, evidence and gap; training follow-up is tied to observed competence needs.

[ ] Claims are traceable to supplied or cited evidence.

[ ] Assumptions, uncertainty and missing evidence are labelled.

[ ] Owners, due dates and next decisions are included where applicable.

[ ] Legal/compliance interpretations needing competent advice are escalated rather than guessed.

[ ] A peer challenge and response are recorded.

Troubleshooting

- The team cannot agree on a score: return to the criterion anchors and record each evidence-based assumption before averaging or escalating.
- Evidence is missing: do not invent it; record an evidence gap, risk implication, owner and retrieval action.
- A legal requirement is unclear: record the current authoritative source and escalate to a competent adviser or authority channel.
- A conclusion feels obvious: test an alternative explanation and state what evidence would falsify the current view.

Facilitator debrief

Ask each team to defend one priority, control or conclusion using objective evidence. Probe what would change the decision, then connect the work to PDCA and the relevant EMS process rather than clause memorisation.

Files in this activity pack

- README.md and README.pdf - detailed learner procedure.
- Activity-A7-Scenario-and-Checklist.pdf - printable scenario and completion checklist.
- Activity-A7-Workbook.xlsx - editable analysis and submission workbook.

Source and usage note

This fictional training activity paraphrases ISO 14001 concepts and is not legal advice or a reproduction of the standard. Use the organisation's licensed ISO 14001:2026 copy and current authoritative Singapore sources for formal interpretation.